

ET617 Learning Platform

Intelligent question generation, real-time analytics, and personalised feedback powered by Bloom's Taxonomy and LLM technology

Group 4





The Challenge Facing Modern Educators

Time-Intensive Question Design

Instructors spend countless hours crafting higher-order thinking questions that truly assess deep learning, often at the expense of other teaching priorities.

Limited Real-Time Visibility

Student misconceptions remain hidden until summative assessments, making timely intervention nearly impossible and leaving gaps unaddressed.

Generic Feedback Loops

Students receive standardised responses that fail to target their specific mistakes, resulting in repeated errors and slower learning progression.

A Comprehensive Solution for Assessment Excellence

ET617 combines intelligent question generation, live assessment delivery, and LLM-powered analytics to transform how instructors teach and students learn.



Instructor Input

Select content and learning objectives



Question Generation

Bloom-aligned questions created automatically



Live Assessment

Quizzes and polls delivered in real-time



Student Responses

Answers captured and analysed instantly



Insights & Feedback

Personalised guidance and instructor coaching

Target Users & Their Needs

Primary: Teachers

Instructors need efficient tools to manage courses, organize assignments, track grades, and provide timely feedback—all without overwhelming administrative overhead.

- Manage multiple courses efficiently
- Create and organize assignments with ease
- Track student progress at a glance
- Provide constructive, timely feedback

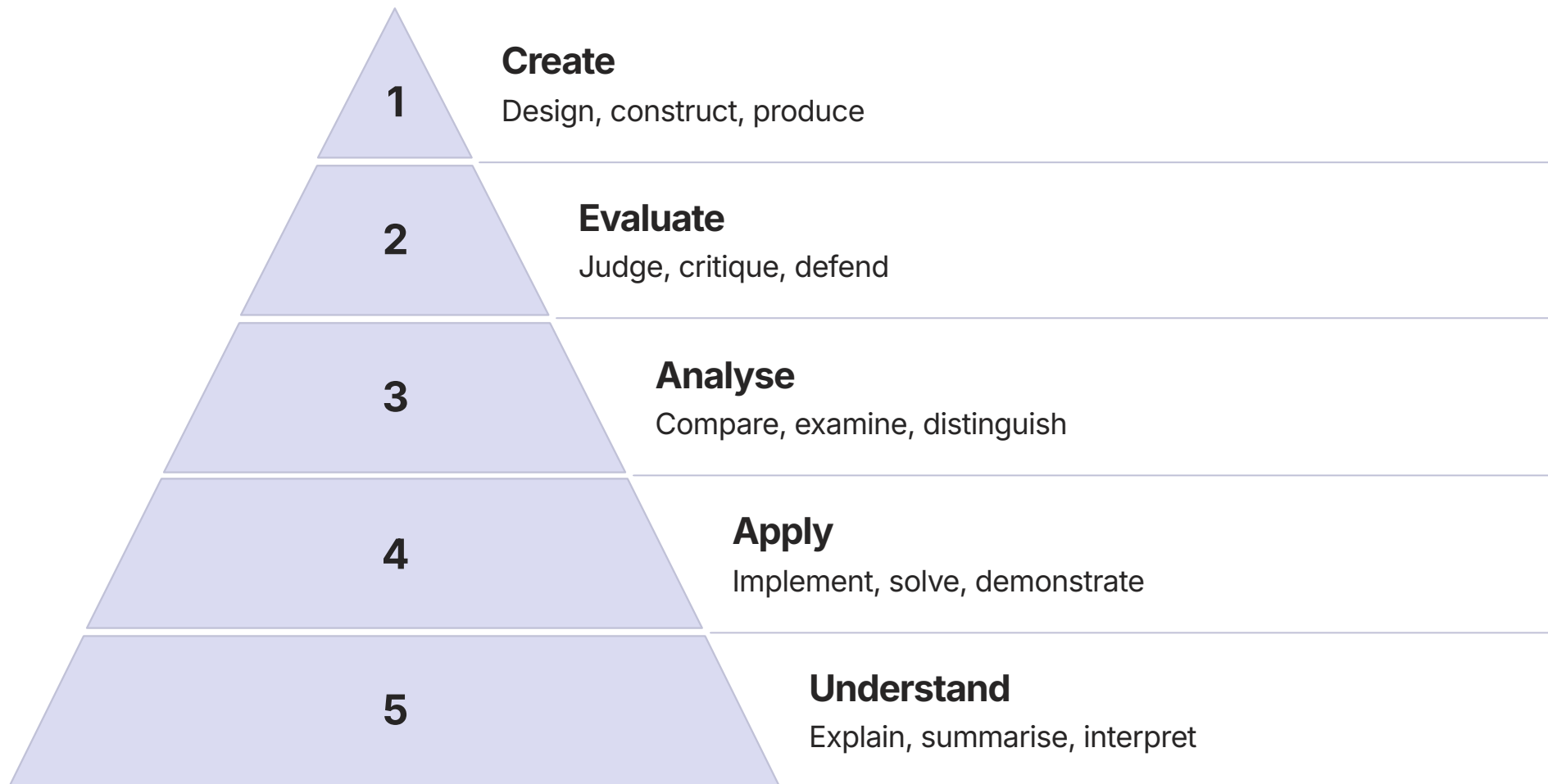
Secondary: University and High school Students

Learners need easy access to course materials, transparent submission workflows, and clear visibility into their own performance and feedback.

- Access course materials anytime, anywhere
- Submit work with confidence
- View grades and feedback transparently
- Track personal learning progress effectively

Bloom's Taxonomy: The Foundation of Quality Assessment

Our platform maps every question to Bloom's cognitive levels, ensuring assessments measure genuine understanding rather than mere recall.



Higher-order thinking skills at the top drive deeper learning and better prepare students for professional practice and research.

Intelligent Question Generation

How It Works

01

Select Content

Instructor identifies topic and target Bloom level

or

Add PDF context and generate questions

02

LLM Generation

System creates multiple question variants

03

Review & Approve

Instructor curates and refines questions

04

Deploy

Questions added to quiz bank

Question Transformation Example



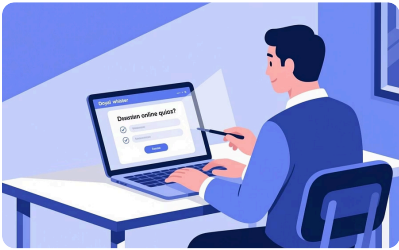
Remember Level: "What is photosynthesis?"

↓ Elevated to ↓



Analyse Level: "Compare the efficiency of C3 and C4 photosynthesis pathways in tropical versus temperate climates. What environmental factors account for these differences?"

Seamless Quiz and Poll Delivery



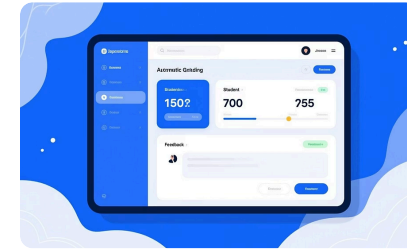
Create Assessments

Build quizzes and polls with drag-and-drop simplicity for live or asynchronous delivery



Live Monitoring

Track participation, correctness, and response times as students engage in real-time



Instant Results

Auto-grading delivers immediate scores whilst students join sessions effortlessly via code



Personalised Student Feedback That Drives Learning

Targeted Remediation

Each student receives feedback specifically addressing their misconceptions, not generic responses. The system identifies error patterns and delivers:

- Conceptual explanations tailored to the mistake
- Worked examples demonstrating correct approaches
- Practice questions targeting the knowledge gap
- Links to relevant course materials and resources

Measurable Impact

3x

Faster mastery

Students close knowledge gaps more quickly

50+%

Reduced errors

Fewer repeated mistakes on subsequent assessments

LLM-Powered Instructor Analytics and Coaching

Transform raw assessment data into actionable teaching strategies with AI-driven insights that guide instructional decisions.

1 Class-Level Summaries

Aggregate performance metrics, question difficulty analysis, and discrimination indices reveal which items effectively differentiate learners

2 Misconception Patterns

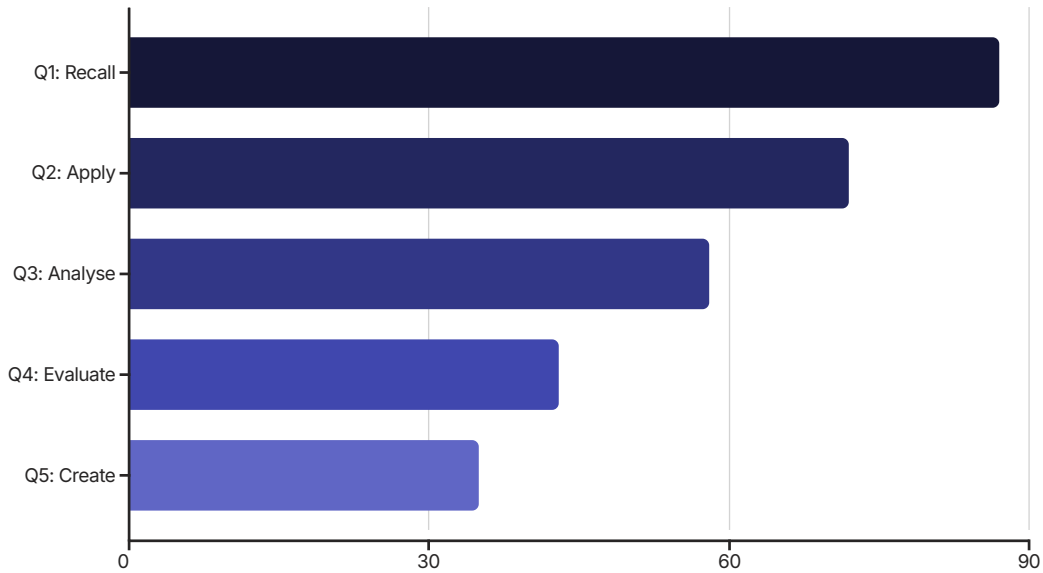
Word clouds and error clustering visualise common student mistakes, helping instructors identify concepts requiring re-teaching

3 Coaching Recommendations

AI suggests specific instructional strategies, examples, and next steps based on class performance patterns and learning science

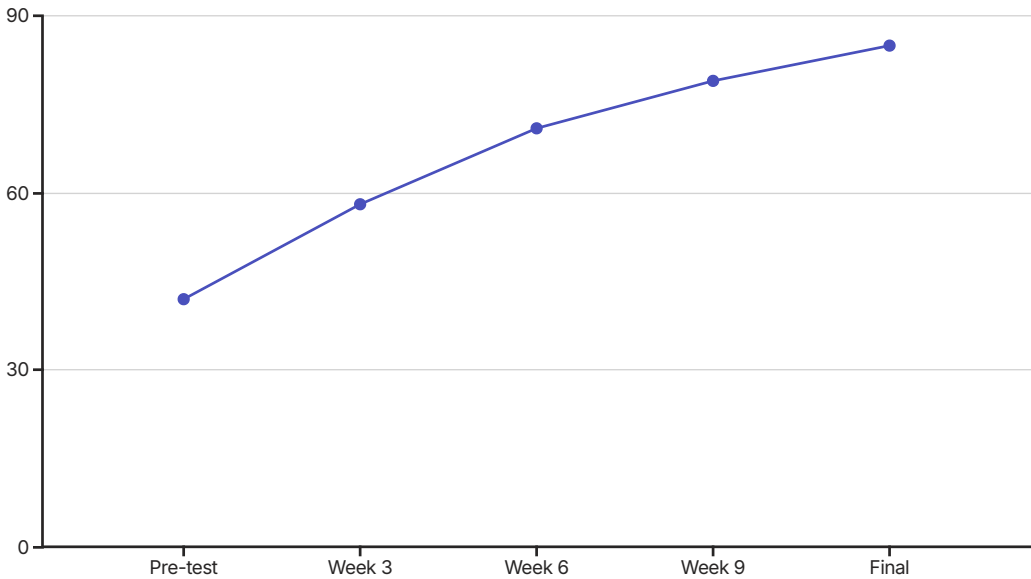
Rich Analytics for Evidence-Based Teaching

Question Performance



Correctness distributions reveal which cognitive levels require additional instructional support.

Learning Gains Over Time



Track aggregate learning progression and measure the effectiveness of interventions across multiple time points.

Secure, Scalable, and Ready for Your Institution

Technical Architecture



Frontend

React + HTML + Tailwind CSS



Backend

Django + PostgreSQL (prod) / SQLite (dev)



LLM Integration

Groq API + Langchain



Security

Auth0

Privacy & Compliance

We prioritise data protection and transparency:

- Pseudonymised student identifiers in LLM prompts
- Role-based access controls for instructors and administrators
- Complete audit trails for all LLM interactions
- Minimal personally identifiable information shared externally
- Transparent logging policies available to institutions

Roadmap: IRT-based item analysis, adaptive quiz assembly, LMS/LTI integrations, expanded remediation libraries, and spaced repetition scheduling.

Key Takeaways & Next Steps

"Design is not just what it looks like—it's how it works." – Steve Jobs

User Empathy Drives Great Design

Understanding your users deeply—their frustrations, goals, and workflows—is the foundation of meaningful design solutions.

Iteration Beats Perfection

Small, frequent cycles of feedback and improvement compound into exceptional results. Done is better than perfect.

PBL Bridges Theory and Practice

Project-based learning forced us to apply design principles, educational theory, and teamwork skills in authentic contexts.

Future Enhancements

- AI-assisted grading
- Chat integration
- Gamified dashboards
- Advanced analytics

[Explore the Live Platform](#)